Lecture_6

2025-09-27

R Markdown

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Working with GGPlot2:

```
install.packages("ggplot2")
library(ggplot2)

View(mpg)
```

Layering ggplot Specifications:

```
myPlot <- ggplot(mpg)
```

- ggplot is a function from the ggplot2 library in R.
- mpg is a built-in dataset about car fuel economy (miles per gallon, car type, engine size, etc.).
- The code line says:"Create a new plot object using the mpg dataset and save it in myPlot."
- At this point, the plot is empty, you've just told R which data to use. You still need to add layers (like + geom_point() for scatter plot, + geom_bar() for bar chart, etc.) to actually see something.

```
# The aesthetic
myPlot <- myPlot + aes(x=displ,y=hwy)
```

- aes(...) means aesthetics. It tells ggplot which columns of the dataset go on the x-axis and y-axis.
- x = displ : Put engine size (displacement) on the x-axis.
- y = hwy : Put highway miles per gallon on the y-axis.
- So now: You told ggplot, "use engine size vs. highway fuel efficiency for the plot."
- The plot is still not visible because you haven't chosen how to draw it (points, bars, lines, etc.). That's the next step:

```
# The geometry
myPlot <- myPlot + geom_point()
```

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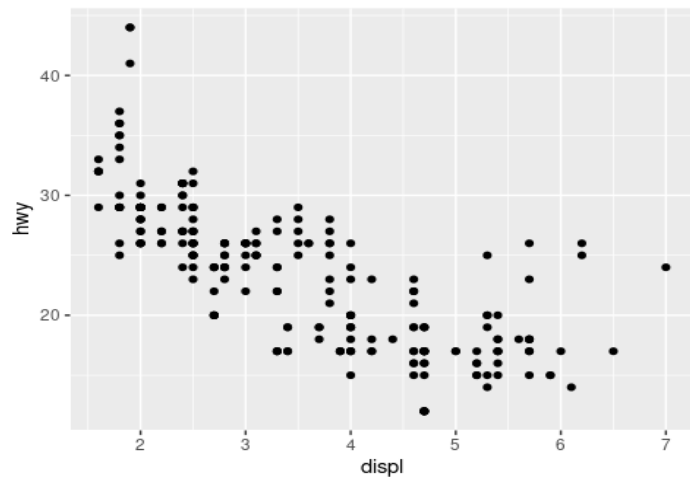
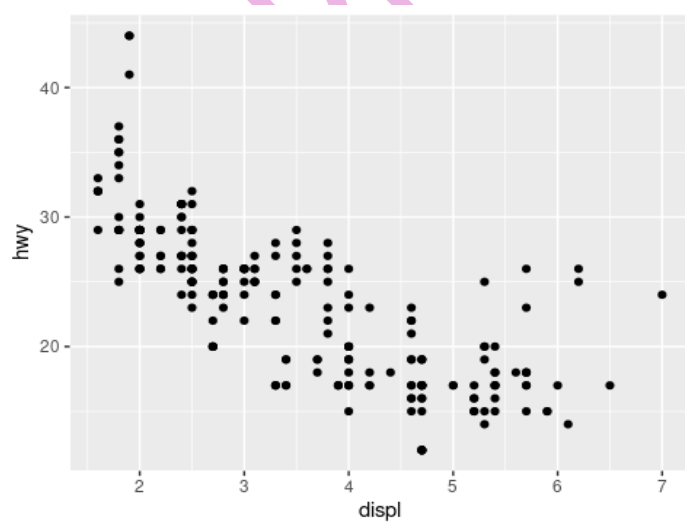
- geom_point () means: draw the data as points. This makes a scatter plot.
- Since you already set x = displ and y = hwy, ggplot will now plot engine size on the x-axis and highway mpg on the y-axis.
- Each dot in the plot = one car from the mpg dataset.

```
myPlot
```



- You see the scatter plot: larger engine size usually means lower highway mpg (dots go downward as displ increases).

```
ggplot(mpg) + aes(x = displ, y = hwy) + geom_point()
```



The above line is the short version of what you built step by step before. Here's what happens in order:

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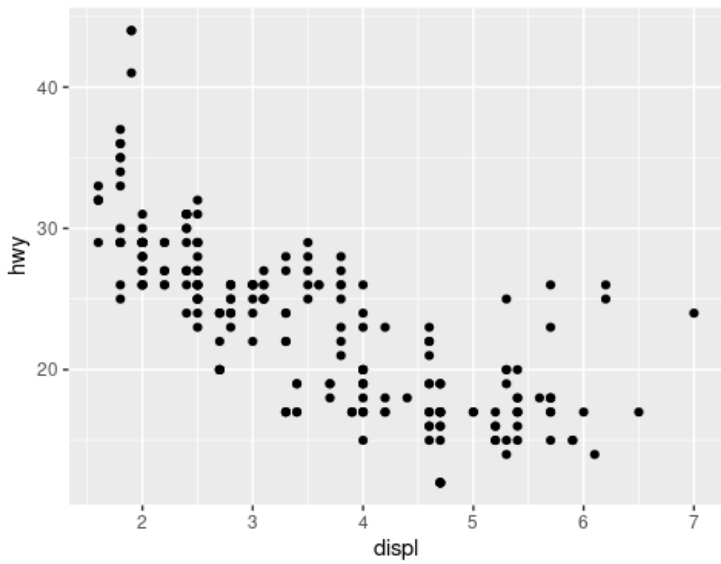
- ggplot(mpg): Use the mpg dataset.
- aes(x = displ, y = hwy) : Map engine size to the x-axis and highway mpg to the y-axis.
- geom_point () : Draw the data as points, gives you a scatter plot.
- **The result:** a scatter plot where each dot = one car, showing the relationship between engine size and fuel efficiency (bigger engines → usually lower mpg).

Using Tidyverse Style

```
library(tidyverse)

mpg %>%
  ggplot() +
     aes(x=displ,y=hwy) +
     geom_point()
```



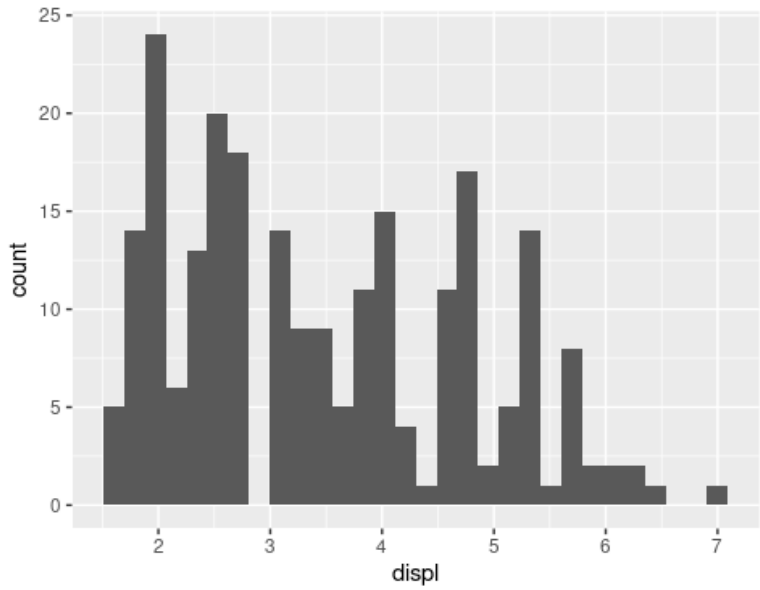
- mpg %>% : the pipe (%>%) sends the mpg dataset into ggplot(). It's just another way of writing ggplot(mpg).
- ggplot() : Start a new plot with the mpg data.
- aes(x = displ, y = hwy) : Map engine size to the x-axis and highway mpg to the y-axis.
- geom_point() : Draw the data as dots (scatter plot).

Univariate Display

```
mpg %>%
  ggplot() +
```

```
  aes(x=displ)+
  geom_histogram()
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



- library(tidyverse) : Loads ggplot2 (and other helpful packages).
- mpg %>% ggplot() : Start a plot with the mpg dataset.
- aes(x = displ): Put engine size (displacement) on the x-axis.
- geom_histogram(): Make a histogram, which shows how many cars fall into different engine size ranges (like 2.0–2.5, 3.0–3.5, etc.).
- The result: a bar-like plot where each bar = number of cars with engine size in that range.
- By default, geom_histogram() picks its own bin width (the size of those ranges). Sometimes that's too wide or too narrow. You can control it with:

Univariate Display: Control Binwidth

```
mpg %>%
  ggplot() +
  aes(x=displ)+
  geom_histogram(binwidth=1)
```

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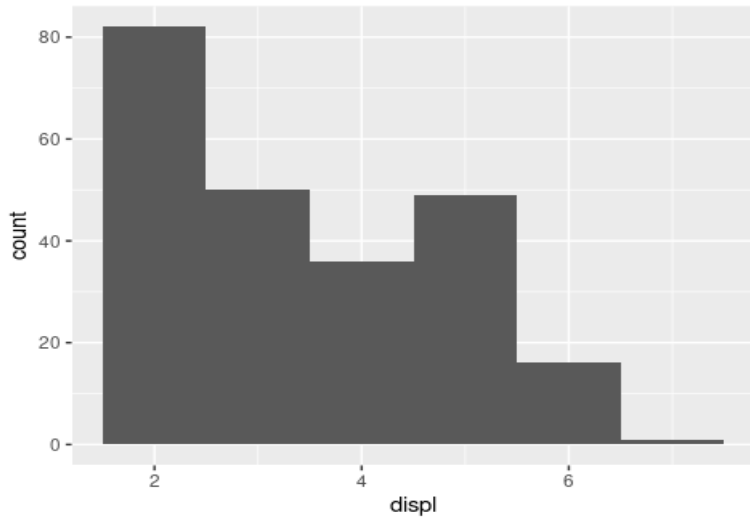
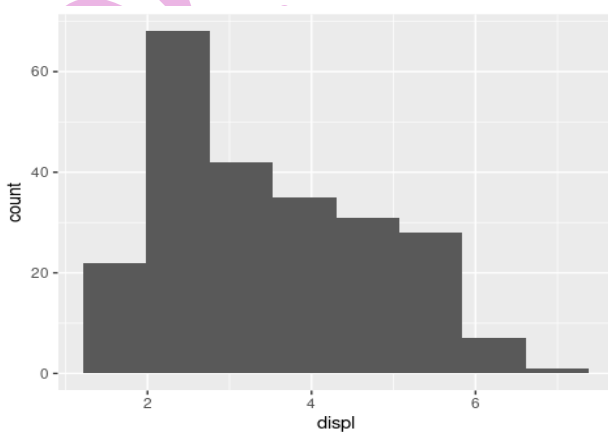


- binwidth = 1 : Each bar covers an engine size range of 1 liter. For example: 1–2 L, 2–3 L, 3–4 L, etc.
- This makes the histogram easier to read because you control how “chunky” or “smooth” the bars are.
- The result: a histogram showing how many cars fall into each 1-liter engine size group.
- If you make binwidth smaller (like 0.2), you’ll see more bars (more detail). If you make it larger (like 2), you’ll see fewer bars (less detail, more summary).

Univariate Display: Set Bin Count

```
mpg %>%
  ggplot() +
  aes(x=displ)+
  geom_histogram(bins=8)
```



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- bins = 8 : ggplot will cut the range of engine sizes into 8 equal parts.
- Each bar shows how many cars fall into that slice of engine sizes. The difference from binwidth is:

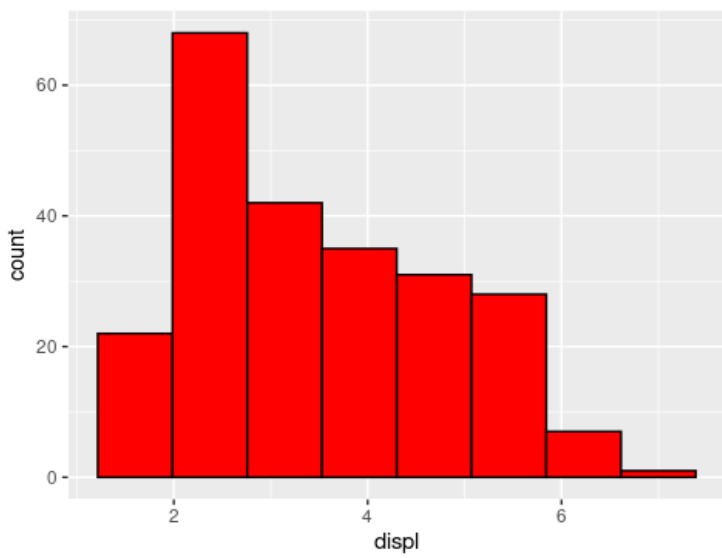
binwidth = you choose how wide each bar is (e.g., 1 liter wide).

bins = you choose how many bars total (e.g., 8 bars across the whole range).

- **The result**: a histogram with exactly 8 bars summarizing the distribution of engine sizes.
- **Tip:** Both binwidth and bins can control the look of a histogram, but usually you set one or the other (not both).

Change Colors

```
mpg %>%
  ggplot() +
  aes(x=displ)+
  geom_histogram(bins=8,
      fill="red",col="black")
```

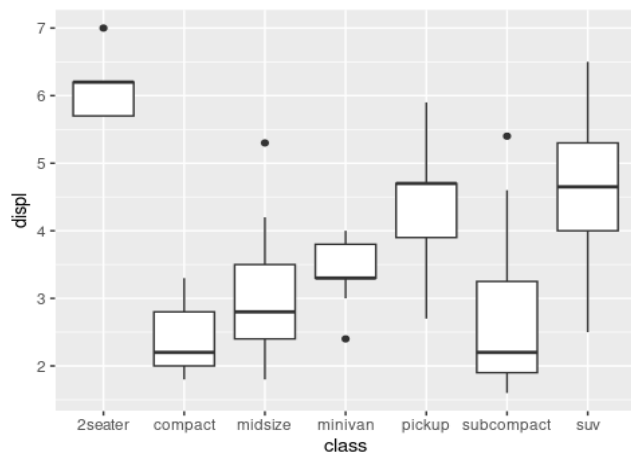


- bins = 8 :Split engine size into 8 groups.
- fill = “red” : The inside of each bar will be red.
- col = “black” : The outline (border) of each bar will be black.
- **The result**: a histogram with 8 red bars outlined in black, showing the distribution of car engine sizes.

- Quick tip: you can use any R color name (like “blue”, “green”, “purple”) or even hex codes (like “#FF5733” for custom shades).

Boxplot

```
mpg %>%
  ggplot() + aes(x=class,y=displ) +
    geom_boxplot()
```



- This code creates a boxplot that shows how big the engine is for different types of cars.
- Here’s the breakdown:

❖ **mpg %>% ggplot ():** It starts the boxplot using the car data (mpg).

❖ **aes(x=class, y=displ):** It puts the type of car (like SUV, compact, etc. class) along the X-axis. It puts the engine size in liters (displ for displacement) up the Y-axis.

❖ **geom_boxplot():** It draws the actual boxplots. Each boxplot shows the range of engine sizes for a specific type of car, including the average, the smallest, the largest, and any unusual engine sizes (outliers).

In short, it compares the engine sizes of different categories of cars.

More Scatter Plots: Adding Color

```
mpg %>%
  ggplot() +
  aes(x=displ,
        y=hwy) +
  geom_point(
       aes(color = cty))
```

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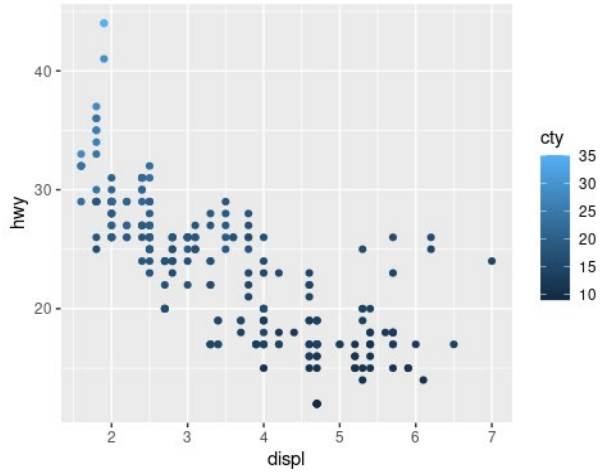

- This code makes a dot chart (a scatterplot) that looks at three things about cars all at once.
- mpg %>% ggplot() + aes(x=displ, y=hwy): It sets up the basic chart: It puts the engine size (displ) across the X-axis. It puts the highway gas mileage (hwy) up the Y-axis.
- This shows how engine size relates to highway efficiency. (Bigger engines usually mean worse highway mileage).
- geom_point (aes(color = cty)): It draws the dots (cars) and makes the color of each dot represent the city gas mileage (cty).
- In simple terms, you get a chart where: You see the trade-off between engine size (left-to-right) and highway mileage (bottom-to-top). The better the city mileage, the more intense or different the color of the dot will be. It lets you quickly see if cars that are good on the highway are also good in the city, based on their engine size.

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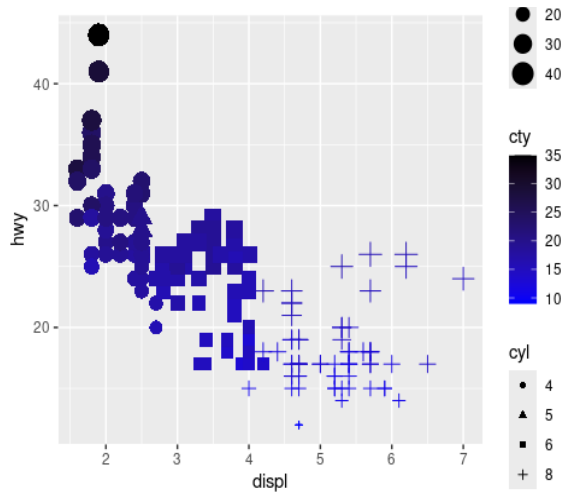
Adding More Attributes

```
mpg %>%
  mutate(cyl=as.factor(cyl)) %>%
  ggplot() +
     aes(x=displ,y=hwy) +
     geom_point(aes(color = cty,
           shape=cyl, size=hwy)) +
           scale_color_gradient(
                      low = "blue",
                      high = "black")
```



- This code creates a highly detailed dot chart (scatterplot) that compares engine size and fuel efficiency for different cars, using four different visual features at once.

  Here's what the chart shows:

- **Basic Relationship (X and Y axes):** The engine size in liters (displ) is shown across the X-axis. The highway gas mileage in MPG (hwy) is shown up the Y-axis.

- **Visual Details (Aesthetics):**

  Dot Size (size=hwy): The bigger the dot, the better the highway gas mileage.

  Dot Color (color=cty): The color tells you the city gas mileage. Blue means low city MPG. Black means high city MPG.

  Dot Shape (shape=cyl): The shape of the dot tells you the number of cylinders (4-cyl, 6-cyl, 8-cyl, etc.). This is why we used mutate(), to treat the number of cylinders as distinct groups for the shapes.

What the Chart Tells You:

Big picture: Cars with small engines are usually high and blue (good highway MPG, low city MPG).

Specific cars: You can spot a black square (high city MPG, 6-cylinder) and see if its dot is high (good highway MPG) or low (bad highway MPG).

Dual Scatterplot with Fitted Line

```
mpg %>%
  ggplot() +
  aes(x=cty,y=hwy) +
  geom_point() +
  facet_wrap(~year) +
  geom_smooth(method="lm")
```

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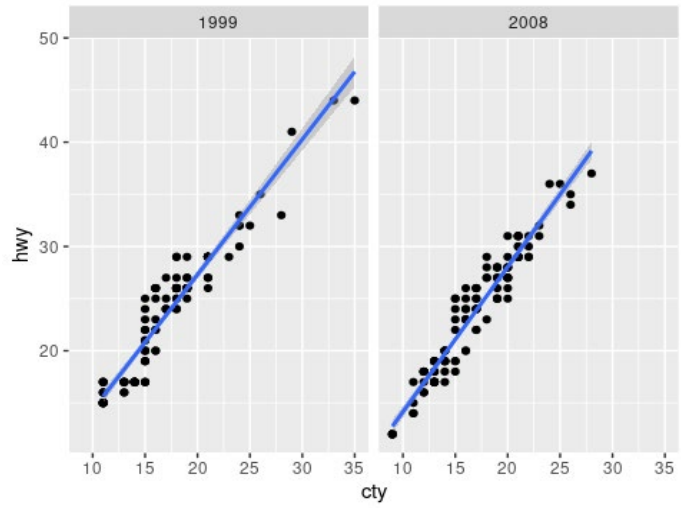
```
## `geom_smooth()` using formula = 'y ~ x'
```



- This code makes two separate charts that compare a car's city gas mileage to its highway gas mileage, one chart for each year of the car models.

Here's the breakdown:

- **mpg %>% ggplot() + aes(x=cty, y=hwy):** It sets up the main comparison: It puts City MPG (cty) across the X-axis, and puts Highway MPG (hwy) up the Y-axis.
- **geom_point():** It draws a dot for every car, showing its specific combination of city and highway mileage.
- **facet_wrap(~year):** This is the key part! It tells the program to split the data and create a separate small plot for each value in the year column (1999 and 2008). You get two distinct graphs side-by-side.
- **geom_smooth(method="lm"):** On both of those small plots, it draws a straight line that best fits through the dots. This line (a linear model or "lm") shows the general trend or relationship between city and highway mileage for that specific year.
- **In simple terms**: You get two charts, one for 1999 cars and one for 2008 cars, each showing the dots for all the cars from that year and a clear line that summarizes how well city mileage predicts highway mileage in that year.

Bar Charts

```
mpg %>%
  group_by(class) %>%
  summarize(hwy=mean(hwy),
            displ=mean(displ)) %>%
  ggplot(aes(x = class, y=hwy))  +
```

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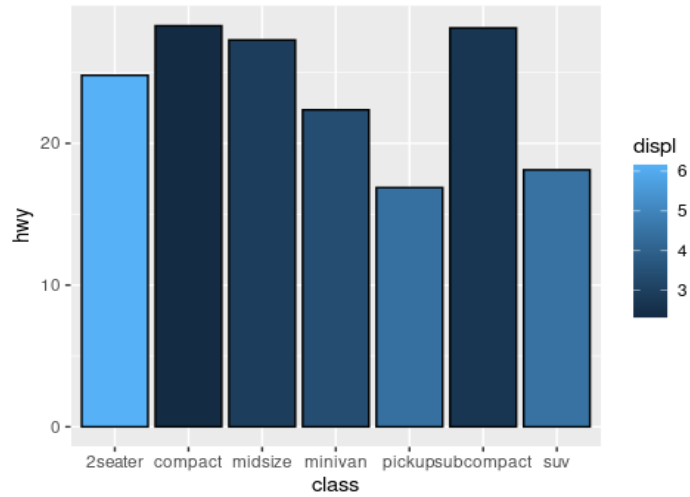
```
  geom_col(color="black",
      aes(fill=displ))
```



This code doesn't plot individual cars; instead, it first calculates the average for each car type, and then makes a chart based on those averages.

Here's the step-by-step simple explanation:

1. **Calculate the Averages (Data Preparation) mpg %>% group_by(class)**: It takes all the car data (mpg) and puts all the similar car types (class) into separate groups (like grouping all SUVs together, all compacts together, etc.).

2. **summarize(hwy=mean(hwy), displ=mean(displ)):** For each of those car type groups, it calculates two averages:
   a. The average highway gas mileage (hwy).
   b. The average engine size (displ).

3. **ggplot (aes(x = class, y=hwy)):** It starts the plot: The type of car (class) goes along the X-axis. The average highway gas mileage (hwy) goes up the Y-axis.

4. **geom_col(color="black", aes(fill=displ)):** It draws bars for each car type: The height of the bar shows the average highway MPG. The color inside the bar (the fill) shows the average engine size (displ). Smaller engines will have one color, and bigger engines will have a different color on the scale.

In simple words: You get a bar chart that shows the average highway fuel efficiency for each type of car. The taller the bar, the better the gas mileage. Plus, the color of the bar gives you a visual hint about the average engine size of that car type.

Bar Charts: Rotate Text

```
mpg %>%
  group_by(class) %>%
```

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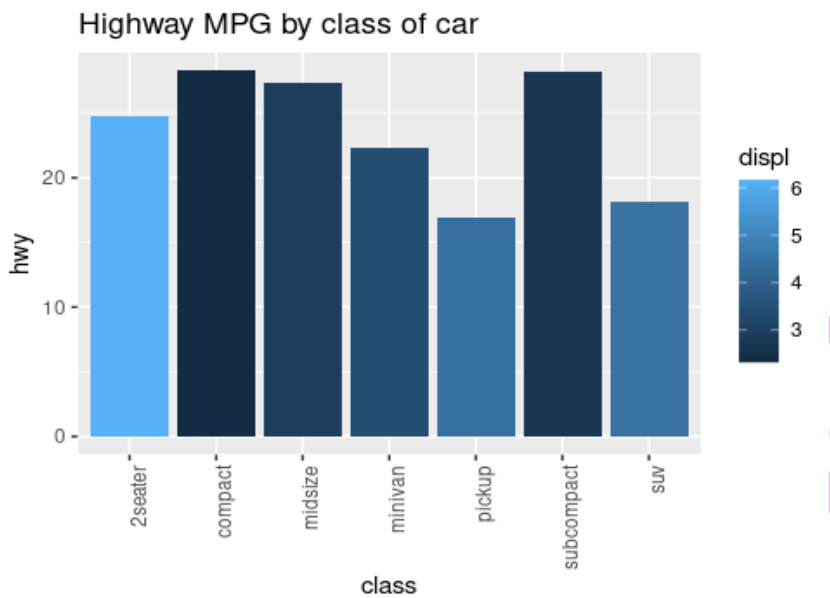
```
  summarize(hwy=mean(hwy), displ=mean(displ)) %>%
  ggplot(aes(x = class, y=hwy))  +
      geom_col(aes(fill=displ)) +
      theme(axis.text.x =
           element_text(angle = 90, hjust = 1)) +
      ggtitle("Highway MPG by class of car")
```



This code creates a bar chart showing the average fuel economy for different types of cars, and it adds some final touches to make the chart easy to read and understand.

Here is the simple, step-by-step explanation:

1. **Calculate the Averages (Data Preparation):**
   mpg %>% group_by(class) %>% summarize(...): This step first groups all the cars by their type (like SUV, compact, etc.) and then calculates two averages for each type:

a. The average highway gas mileage (hwy).
b. The average engine size (displ).

2. **Draw the Chart (Plotting)**
   ggplot(aes(x = class, y=hwy)) + geom_col(aes(fill=displ)): This draws the bar chart:

- The Type of Car (class) goes along the X-axis.
- The Average Highway MPG (hwy) determines the height of the bar.
- The Average Engine Size (displ) determines the color inside the bar (the fill).

3. **Polish the Chart (Final Touches):**

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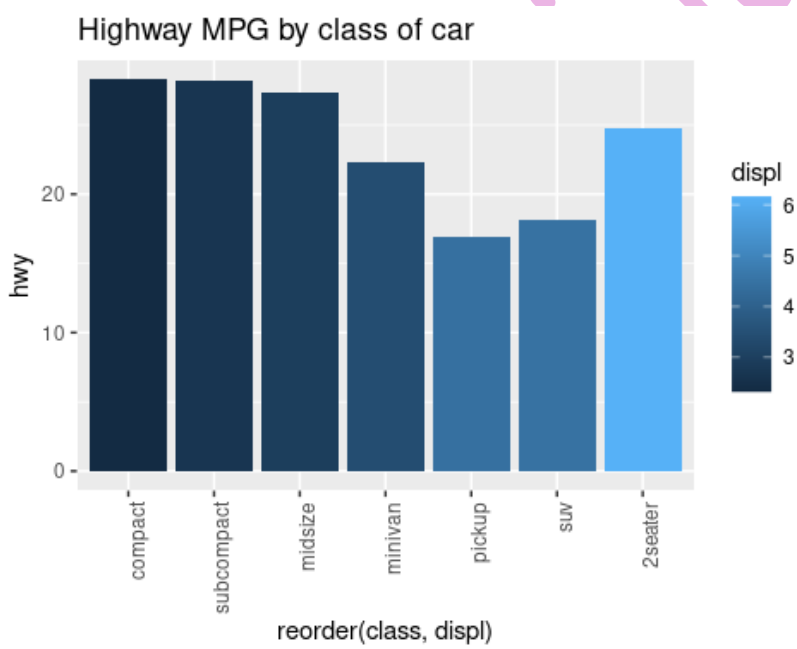
theme (axis.text.x = element_text(angle = 90, hjust = 1)): This is a cleanup step. Since the car type names (like "subcompact") can be long, this command rotates the names on the bottom axis by 90 degrees, so they don't overlap and are easy to read.

ggtitle("Highway MPG by class of car"): This simply adds a main title to the top of the chart so everyone knows exactly what they are looking at.

In simple words: You get a bar chart showing the average highway mileage for each car type, with the bars colored by their average engine size. The car type names on the bottom are rotated so they fit nicely, and the chart has a clear title.

Bar Charts: reorder the columns:

```
mpg %>%
  group_by(class) %>%
  summarize(hwy=mean(hwy), displ=mean(displ))  %>%
  ggplot(aes(x = reorder(class, displ),y=hwy))  +
  geom_col(aes(fill=displ)) +
  theme(axis.text.x =
           element_text(angle = 90, hjust = 1)) +
  ggtitle("Highway MPG by class of car")
```



This code creates the exact same bar chart as before, but with one important change: it sorts the car types along the bottom axis based on their average engine size.

Here's the simple, **step-by-step explanation:**

1. **Calculate the Averages**

mpg %>% group_by(class) %>% summarize(...): This part still does the initial math. It groups cars by their type (class) and calculates the average highway MPG (hwy) and the average engine size (displ) for each group.

2. **Draw and Sort the Chart**

ggplot(aes(x = reorder(class, displ), y=hwy)): This is the main change! Instead of just listing the car types (class) randomly or alphabetically on the X-axis, the reorder(class, displ) function sorts the car types by their average engine size (displ).

geom_col(aes(fill=displ)): This draws the bars. The height shows the average highway MPG, and the color shows the average engine size (big engines have one color, small engines another).

3. **Polish the Chart theme(...):** Rotates the car type names on the bottom by 90 degrees, so they don't overlap.

ggtitle(...): Adds a title to the chart.

In simple words: You get a bar chart that shows the average fuel economy for each car type, but the car types are neatly arranged from the smallest average engine size on the left to the largest average engine size on the right. This makes it much easier to spot the relationship between engine size and fuel efficiency.

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